

Total No. of Questions : 8]

[Total No. of Printed Pages : 3

Roll No

FT-7004(2)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)
Operations Research and Supply Chain

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Define Linear Programming. What are the advantages and limitations of LP problem?
b) A business man has the option of investing his money in two plans. Plan A guarantees that each rupee invested will earn seventy paise a year hence, while plan B guarantees that each rupee invested will earn two rupees two years hence. In plan B, only investments for periods that are multiples of two years are allowed. The problem is how he should invest ten thousand rupees in order to maximize the earnings at the end of three years. Formulate this problem as a linear programming model.
2. a) Explain the role of Handling systems in SCM.
b) Explain the factors influencing the decision making process of outsourcers.
3. a) Define a queue.

FT-7004(2)-CBGS

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[2]

- b) In a bank cheques are cashed at a single teller counter. Customers arrive at the counter in a Poisson manner at an average of 20 customers per hour. The teller takes on an average 2 minutes to cash cheque. The service time is exponentially distributed.
- Calculate the percentage of time the teller is busy.
 - Calculate the average time a customer is expected to wait.
4. a) What is forecasting? Explain risk management forecasting in detail.
- b) Define ERP. Also explain the interface modules of ERP with SCM.
5. Solve the given transportation problem by VAM and North West corner method

Source	Destination				
	1	2	3	4	Supply
1	15	18	12	16	50
2	14	19	20	14	30
3	13	16	13	17	40
Demand	30	30	25	35	

6. a) What do you understand by decision tree Analysis? Discuss decision under certainty.
- b) Explain Hurwicz Criteria. What do you mean by meta-nearestic algorithms?

[3]

7. a) Explain VED analysis used in inventory control. Write advantages and limitations.
b) Write short note on following:
 - i) Single Server Model (M/M/L)
 - ii) Multi Server Model (M/M/S)

8. Write short notes on followings(any four) :
 - i) Saddle point
 - ii) Monte Carlo simulation
 - iii) Multiple server model
 - iv) Vogel's method
 - v) MRP calculation
